1. SITE SELECTION & LAND REQUIREMENTS

1.1 Site Locations (Per Project Proposal)

SITE	LOCATION	COORDINATES	ELEVATION	LAND AVAILABLE
LETPANTAW	Letpantaw Village, Magway Township	TBD	~100 m	120 acres
MYINSAI	Myinsai Village, Magway Township	TBD	~100 m	300 acres
KANSWEI	Kanswei Village, Magway Township	TBD	~100 m	240 acres

Note: All three sites are within approximately 5-mile radius of each other.

1.2 Land Requirements

Per 3.2 MWac Block (Fixed-Tilt):

- PV array area: ~1.8 hectares (4.4 acres)
- Roads, drainage, spacing: ~0.7 hectares (1.7 acres)
- Total per block: ~2.5 hectares (6.2 acres)

Site-Level Land Budget:

Site	PV Blocks	PV Area	Substation	BESS	Roads/Buffer	Total Required	Available
Letpantaw	10	25 ha (62 acres)	3 ha (7 acres)	0.5 ha (1.2 acres)	15 ha (37 acres)	43.5 ha (107 acres)	49 ha (120 acres) ✓
Myinsai	16	40 ha (99 acres)	2 ha (5 acres)	0.8 ha (2 acres)	20 ha (49 acres)	62.8 ha (155 acres)	300 acres✓
Kanswei	16	40 ha (99 acres)	2 ha (5 acres)	0.8 ha (2 acres)	20 ha (49 acres)	62.8 ha (155 acres)	97 ha (240 acres) ✓

Land density: ~0.31 MWac/acre (matches 120 MW project's 0.306 MWac/acre)

1.3 Site Conditions & Assumptions

Climate (Magway Dry Zone):

- Annual GHI: ~1,900-2,000 kWh/m²/year
- Ambient temperature range: 5°C to 45°C
- Monsoon season: May-October
- Soiling: Moderate to high (dry season dust)

Geotechnical (Assumed, pending site investigation):
• Soil type: Sandy clay / laterite
• Bearing capacity: 100-150 kPa (suitable for shallow foundations)
• Groundwater: >5 m depth
• Seismic zone: Zone III-IV per Myanmar code (0.3-0.4g PGA)
Topography:
• Terrain: Relatively flat (<5% slope)
• Vegetation: Sparse scrubland
• Drainage: Natural surface runoff to be managed

2. PV SYSTEM DESIGN

2.1 Module Specification

Selected Module: 635 Wp Bifacial Monocrystalline Silicon

Key Specifications:
• Rated power (Pmax): 635 Wp (STC)
• Efficiency: 23.51%
• Open-circuit voltage (Voc): 49.68 V
• MPP voltage (Vmpp): 41.16 V
• MPP current (Impp): 15.43 A
• Short-circuit current (Isc): 16.26 A
• Bifacial factor: 70%
• Temperature coefficient (Pmax): -0.29%/°C
• Dimensions: 2384 x 1303 x 35 mm
• Weight: 34.5 kg
• Technology: Monocrystalline PERC with bifacial cells
• Certifications: IEC 61215, IEC 61730, UL 1703

2.2 String Configuration

String Design: 28 modules in series

String power (STC): 28 × 635 Wp = 17.78 kWp

Voltage Analysis:
• Cold (worst case, -5°C): String Voc = 1,495 V, String Vmpp = 1,237 V
• STC (25°C): String Voc = 1,391 V, String Vmpp = 1,152 V
• Hot (operating, 70°C): String Voc = 1,231 V, String Vmpp = 1,023 V

Safety Margin:
• Max system voltage: 1,500 Vdc
• Cold Voc: 1,495 V → 5 V margin (acceptable for Myanmar climate)
• Inverter MPP range: 500-1,500 V → String operates 1,023-1,237 V ✓

Current Analysis:
• String Impp: 15.43 A
• String Isc: 16.26 A
• Max strings per MPPT: 40 A ÷ 16.26 A = 2.46 → 2 strings per MPPT ✓

2.3 Inverter Specification

Selected Inverter: Sungrow SG320HX Multi-MPPT String Inverter

Key Specifications:
• Model: SG320HX
• AC Output Power: 320 kW @ 40°C
• Max DC Input Voltage: 1,500 V
• Startup Voltage: 550 V
• MPP Voltage Range: 500-1,500 V
• Nominal DC Voltage: 1,080 V
• Number of MPPTs: 12 independent
• Strings per MPPT: 2 (max)
• Max Current per MPPT: 40 A
• Max Efficiency: 99.02%
• European Efficiency: 98.8%
• AC Output Voltage: 800 Vac nominal (3-phase)
• Protection: IP66, C5 anti-corrosion, DC/AC Type II SPD
• Cooling: Smart forced air cooling
• Operating Temp Range: -30°C to +60°C

Inverter Loading:
• Strings per inverter: 22
• DC capacity per inverter: 22 × 17.78 kWp = 391.2 kWp
• ILR (DC/AC ratio): 391.2 ÷ 320 = 1.22 ✓

2.4 Block Design (Standardized 3.2 MWac Unit)

Block Configuration:
• Inverters per Block: 10 × SG320HX
• Block AC Rating: 3,200 kW (3.2 MWac)
• Block DC Rating: 3,912 kWp (3.912 MWp)
• ILR: 1.22
• Strings per Block: 220 (22 per inverter × 10)
• Modules per Block: 6,160 (220 strings × 28 modules)
• Block Transformer: 1 × 3.15 MVA or 3.5 MVA, 800 Vac / 33 kV
• Footprint: ~2.5 hectares (6.2 acres)

2.5 Array Mounting & Structure

Mounting System: Fixed-Tilt (South-Facing)

- Tilt Angle: 26° (optimized for Magway latitude ~20°N)
- Azimuth: 180° (due south)
- Module Orientation: Portrait (vertical)
- Modules per Row: 28 (single row)
- Row Spacing (Pitch): 6.5-7.0 m (to avoid shading 9:00-15:00 at winter solstice)
- Ground Clearance: 1.5 m minimum (bottom edge of module)
- Structure Material: Hot-dip galvanized steel (HDG), ASTM A36 or equivalent
- Foundation Type: Driven steel piles or cast-in-place concrete footings
- Pile Depth: 1.5-2.5 m (depending on soil conditions)
- Wind Load: 50 m/s (3-second gust) per Myanmar code
- Seismic Load: 0.3-0.4g PGA (Zone III-IV)
- Design Life: 25 years minimum

Bifacial Considerations:
- Mounting height: 1.5 m minimum for rear-side irradiance
- Ground treatment: Natural soil (albedo ~20-25%) or white gravel (albedo ~40-50%)
- Expected bifacial gain: 5-10% depending on albedo and mounting height

2.6 Portfolio-Wide PV System Quantities

Item	Letpantaw (30 MW)	Myinsai (50 MW)	Kanswei (50 MW)	Total
PV Blocks	10	16	16	42
Inverters (SG320HX)	100	160	160	420
Block Transformers (3.15 MVA)	10	16	16	42
PV Modules (635 Wp)	61,600	98,560	98,560	258,720
Strings	2,200	3,520	3,520	9,240
DC Capacity	39.1 MWp	62.6 MWp	62.6 MWp	164.3 MWp
AC Capacity	32.0 MWac	51.2 MWac	51.2 MWac	134.4 MWac

3. ENERGY STORAGE SYSTEM (ESS)

3.1 BESS Sizing Philosophy

Requirement: 20% of AC capacity for 2 hours (per Myanmar grid code and 120 MW project precedent)

Site	AC Capacity	BESS Power	BESS Energy	Duration
Letpantaw	30 MW	6 MW	12 MWh	2 hours
Myinsai	50 MW	10 MW	20 MWh	2 hours
Kanswei	50 MW	10 MW	20 MWh	2 hours
Total	130 MW	26 MW	52 MWh	2 hours

3.2 Battery Technology

Selected Chemistry: Lithium Iron Phosphate (LFP)

Rationale:
- High cycle life (6,000-8,000 cycles @ 80% DOD)
- Superior thermal stability and safety vs. NMC
- Lower cost per kWh than NMC
- Proven in utility-scale applications
- Matches 120 MW project selection

Battery Cell Specification (Typical):
- Nominal voltage: 3.2 V
- Capacity: 280-314 Ah
- Energy: ~1 kWh per cell
- Cycle life: >6,000 cycles @ 80% DOD, 25°C
- Calendar life: 15-20 years
- Operating temperature: -20°C to +55°C
- Safety: UL 1973, IEC 62619

3.3 BESS Configuration

Containerized System (20-ft or 40-ft ISO containers)

Per 2 MW / 4 MWh Container (Typical):
- Battery racks: 8-12 racks per container
- Cells per rack: ~200-300 cells in series/parallel
- Battery Management System (BMS): Integrated, cell-level monitoring
- HVAC: Air conditioning (maintain 15-30°C)
- Fire suppression: Aerosol or FM-200, automatic detection
- DC voltage: 1,000-1,500 Vdc typical
- Dimensions: 40-ft container (12 × 2.4 × 2.9 m)
- Weight: ~30-35 tons (loaded)

Power Conversion System (PCS):
- Type: Bi-directional inverter (DC ↔ AC)
- Topology: Modular, transformerless or with integrated transformer
- Efficiency: >97% (round-trip DC-AC-DC ~95%)
- Response time: <100 ms (for grid support)
- Grid codes: LVRT, HVRT, frequency response, black start capability

Step-Up Transformer (per BESS unit):
- Rating: 2.5-3.5 MVA
- Voltage: 0.69 kV (or 0.8 kV) / 33 kV
- Type: Dry-type or oil-immersed, pad-mount or containerized
- Impedance: 6-8%

4. SUBSTATION & ELECTRICAL INFRASTRUCTURE

4.1 Voltage Levels & Architecture

Collection Voltage: 33 kV (site-level)
Transmission Voltage: 66 kV (inter-site) and 230 kV (grid export)

Power Flow:
1. PV blocks (800 Vac) → Block transformers → 33 kV site bus
2. BESS (690 Vac) → BESS transformers → 33 kV site bus
3. Myinsai 33 kV → Step-up to 66 kV → Transmission to Letpantaw
4. Kanswei 33 kV → Step-up to 66 kV → Transmission to Letpantaw
5. Letpantaw: 66 kV from satellites → Step-down to 33 kV → Combine with local 33 kV → Step-up to 230 kV → Grid

4.2 Letpantaw Substation (30 MW + Hub)

Function:
- Collect 30 MW local PV + 6 MW BESS
- Receive 50 MW from Myinsai (via 66 kV)
- Receive 50 MW from Kanswei (via 66 kV)
- Aggregate 130 MW total
- Export to grid via 230 kV

33 kV Switchgear Configuration:

Bay #	Function	Equipment	Rating
1	Main Transformer Feeder	VCB + DS + ES	1,600 A
2-11	PV Block Feeders (10)	VCB + DS + ES	200 A each
12	BESS Feeder #1	VCB + DS + ES	200 A
13	BESS Feeder #2	VCB + DS + ES	200 A
14	SVG Bay	VCB + DS + ES	400 A
15	Myinsai 66/33 kV Transformer	VCB + DS + ES	1,250 A
16	Kanswei 66/33 kV Transformer	VCB + DS + ES	1,250 A
17	Bus PT & Metering	VT + Meters	—
18	Auxiliary Transformer	VCB + DS	100 A

Total: 18 bays

Main Equipment:

33 kV Switchgear Specification:
- Type: KYN-40.5 kV metal-enclosed, withdrawable, indoor

• Rated voltage: 40.5 kV
• Rated current: 1,600 A (main bus), 200-1,250 A (feeders)
• Short-circuit rating: 31.5 kA (3 sec) or 40 kA
• Circuit breaker: Vacuum type (VCB), spring-operated

Main Power Transformer (33/230 kV):
• Rating: 150 MVA (ONAN/ONAF)
• Voltage: 33 kV ± 10% / 230 kV
• Vector group: YNd11 (or Dyn11)
• Impedance: 10-12%
• Cooling: ONAN (base) / ONAF (forced air, fans)
• Tap changer: OLTC (On-Load Tap Changer), ±10% in 17 steps
• Standards: IEC 60076

66/33 kV Step-Down Transformers (2 units):
• Rating: 60 MVA each (ONAN)
• Voltage: 66 kV / 33 kV ± 5%
• Vector group: YNyn0 or Dyn11
• Impedance: 8-10%
• Cooling: ONAN
• Tap changer: OLTC, ±5% or ±10%

SVG (Static Var Generator):
• Rating: ±7.5 Mvar (25% of 30 MW local)
• Voltage: 33 kV
• Response time: <30 ms
• Control modes: Voltage regulation, VAR dispatch, power factor control

230 kV Switchyard:
• Configuration: Single bus or breaker-and-a-half (per utility requirement)
• Bays: 1 × Transformer bay + 1 × Line bay (to Taung-Dwin-Gyi 230 kV substation)
• Circuit breakers: 230 kV SF6, 40 kA, 2,000 A
• Area: ~5,000 m²

4.3 Myinsai Substation (50 MW Satellite)

Function:
• Collect 50 MW PV + 10 MW BESS
• Step up to 66 kV
• Transmit to Letpantaw via 66 kV overhead line

33 kV Switchgear Configuration:

Bay #	Function	Equipment	Rating
1	Main Transformer Feeder	VCB + DS + ES	1,600 A

2-17	PV Block Feeders (16)	VCB + DS + ES	200 A each
18	BESS Feeder #1	VCB + DS + ES	300 A
19	BESS Feeder #2	VCB + DS + ES	300 A
20	SVG Bay	VCB + DS + ES	600 A
21	Bus PT & Metering	VT + Meters	—
22	Auxiliary Transformer	VCB + DS	100 A

Total: 22 bays

Main Equipment:
• 33 kV Switchgear: Same specification as Letpantaw (KYN-40.5 kV)
• Step-Up Transformer (33/66 kV): 60 MVA (ONAN), YNd11, OLTC
• SVG: ±12 Mvar (24% of 50 MW)
• BESS: 10 MW / 20 MWh (5 × 2 MW / 4 MWh containers)
• 66 kV Switchyard: Simple 2-bay configuration (transformer + line)

4.4 Kanswei Substation (50 MW Satellite)

Configuration: Identical to Myinsai (for standardization and procurement efficiency)

4.5 Inter-Site Transmission (66 kV Lines)

Myinsai to Letpantaw (66 kV OH Line):
• Length: 2-5 miles (preliminary, pending final routing within ~5-mile site radius)
• Voltage: 66 kV
• Configuration: Single-circuit, 3-phase
• Conductor: ACSR 240 mm² or 300 mm² (e.g. Zebra, Panther)
• Current capacity: ~500 A (adequate for 50 MW)
• Structures: Steel lattice towers or concrete poles, H-frame or single pole
• Span: 150-200 m typical
• Insulation: Porcelain or composite disc insulators, 66 kV class (5-7 discs)
• Ground wire: ACSR 70 mm² or OPGW (for communication)
• Right-of-way: 30-40 m width
• Design criteria: Wind 50 m/s, seismic 0.3-0.4g PGA

Kanswei to Letpantaw (66 kV OH Line):
• Identical specification to Myinsai line
• Length: 2-5 miles (preliminary, pending final routing within ~5-mile site radius)

Line Losses (Estimated):
• 50 MW, 2-5 miles, 240-300 mm² ACSR: ~0.5-1.5% losses
• Annual energy loss: ~0.5-1.5% of transmitted energy

Note: All three sites are within approximately 5-mile radius. Final routing and distances to be confirmed during detailed design.

4.6 Grid Interconnection (230 kV)

Point of Interconnection (POI):
- Taung-Dwin-Gyi 230 kV Primary Substation
- Distance from Letpantaw: 12.93 miles (approximately 20.8 km)
- Voltage: 230 kV
- Connection: Single-circuit or double-circuit 230 kV overhead line

230 kV Transmission Line (Letpantaw to Taung-Dwin-Gyi):
- Length: 12.93 miles (20.8 km)
- Voltage: 230 kV
- Configuration: Single-circuit (or double-circuit if required by utility)
- Conductor: ACSR 400 mm² or larger (e.g. Moose, Zebra)
- Current capacity: ~800 A (adequate for 130 MW)
- Structures: Steel lattice towers, suspension and tension types
- Span: 200-300 m
- Insulation: 230 kV class, 13-15 disc insulators
- Ground wire: OPGW for communication
- Right-of-way: 40-50 m
- Design: Per utility standards

Estimated Losses:
- 130 MW, 20.8 km, 400 mm² ACSR: ~2.5-3.5% losses
- Annual energy loss: ~2.5-3.0% of exported energy

Grid Code Compliance:
- Voltage regulation: ±5% at POI
- Frequency: 50 Hz ± 0.5 Hz
- Power factor: 0.95 leading to 0.95 lagging (adjustable via SVG and inverters)
- Harmonics: THD <5% (voltage), <3% (current)
- LVRT/HVRT: Per Myanmar grid code
- Active/reactive power control: Ramp rate limiting, curtailment capability

5. PERFORMANCE MODELING & ENERGY YIELD

5.1 Solar Resource (Magway)

Location: Magway Township, Magway Region, Myanmar
- Latitude: ~20°N
- Longitude: ~95°E
- Elevation: ~100 m

Solar Resource (Typical Meteorological Year):

- Global Horizontal Irradiance (GHI): 1,900-2,000 kWh/m²/year
- Direct Normal Irradiance (DNI): 1,800-1,900 kWh/m²/year
- Diffuse Horizontal Irradiance (DHI): 600-700 kWh/m²/year
- Global Tilted Irradiance (GTI, 26° tilt): 2,050-2,150 kWh/m²/year

Temperature:
- Annual average: 28°C
- Monthly average: 20°C (December) to 35°C (April)
- Extreme: 5°C (min) to 45°C (max)

Data Source: NASA SSE, Meteonorm, or local weather station

5.2 Loss Budget

Loss Category	Value	Notes
Soiling	3-5%	Dry season dust, monthly cleaning assumed
Module mismatch	1%	Factory tolerance ±3%, actual ~1%
Module quality/LID	1.5%	Light-induced degradation (first year)
Ohmic losses (DC cables)	1.5%	String cables + DC wiring
Inverter efficiency	1.5%	98.5% weighted average efficiency
Ohmic losses (AC cables)	0.5%	LV + MV cables
Transformer losses	1.0%	Block transformers + main transformers
66 kV transmission	0.5-1.5%	Inter-site lines (2-5 miles each)
230 kV transmission	2.5-3.5%	Grid line (12.93 miles / 20.8 km)
Availability	1.0%	Maintenance, grid curtailment
Shading	0.5%	Near-zero (flat site, proper spacing)
Temperature	5-7%	Module temp 50-70°C, Pmax coeff -0.29%/°C
Inverter clipping	1.0%	ILR 1.22, moderate clipping
Other	0.5%	Miscellaneous

Total Losses: 20-26%

Performance Ratio (PR): 74-80% (accounting for longer 230 kV transmission distance)

5.3 Energy Yield Estimate

Annual Energy Production (AEP) - Portfolio:

Method: PVsyst-based calculation

- GTI (26° tilt): 2,100 kWh/m²/year
- DC capacity: 164.3 MWp
- Performance ratio: 77% (mid-range, accounting for transmission losses)
- Bifacial gain: 7%

AEP = 164.3 MWp × 2,100 kWh/m²/kWp × 0.77 × 1.07
AEP ≈ 280,000 MWh/year (at POI - Taung-Dwin-Gyi Substation)

Specific Yield: ~1,705 kWh/kWp/year (at POI)

Capacity Factor: 280,000 MWh ÷ (164.3 MW × 8,760 hrs) = 19.5%

Note: Performance ratio and capacity factor are reduced compared to shorter transmission distances due to the 12.93-mile (20.8 km) 230 kV line losses (~2.5-3.5%).

Per Site (at site boundary, before transmission):

Site	DC Capacity	AEP (MWh/year)	Capacity Factor
Letpantaw	39.1 MWp	70,400	22.5%
Myinsai	62.6 MWp	112,700	22.5%
Kanswei	62.6 MWp	112,700	22.5%
Total (at sites)	164.3 MWp	295,800	22.5%

Energy at POI (after 230 kV line losses): ~280,000 MWh/year

Annual 230 kV transmission loss: ~8,000-10,000 MWh/year (~2.7-3.4% of site generation)

5.4 Performance Degradation

Module Degradation:
- Year 1: 2% (LID + initial)
- Year 2-25: 0.40% per year
- Year 25: 80% of nameplate (per warranty)

25-Year Energy Production:
- Year 1: 295,800 MWh × 0.98 = 289,900 MWh (at sites)
- Year 25: 295,800 × 0.904 = 267,400 MWh (at sites)
- Total 25-year (at sites): ~7,000,000 MWh (7.0 TWh)
- Total 25-year (at POI): ~6,650,000 MWh (6.65 TWh) after transmission losses

7. NEXT STEPS & RECOMMENDATIONS

IMMEDIATE ACTIONS:
1. Confirm Myinsai land acreage from final survey
2. Conduct detailed routing study for 66 kV lines (Myinsai and Kanswei to Letpantaw)
3. Finalize 230 kV transmission line route (12.93 miles to Taung-Dwin-Gyi)
4. Update CapEx estimate to reflect corrected transmission distances

5. Conduct financial analysis with revised energy yield (accounting for transmission losses)

ENGINEERING STUDIES REQUIRED:
• Load flow study for 66 kV and 230 kV systems
• Short-circuit analysis at all voltage levels
• Protection coordination study
• Transmission line design (tower spotting, foundation design)
• Geotechnical investigation at all sites and along transmission routes
• Environmental impact assessment for transmission corridors

PERMITTING & APPROVALS:
• Grid interconnection agreement with utility (at Taung-Dwin-Gyi)
• Right-of-way acquisition for 66 kV lines (2-5 miles each)
• Right-of-way acquisition for 230 kV line (12.93 miles)
• Land use permits for all three sites
• Environmental clearances
• Construction permits

COMMERCIAL CONSIDERATIONS:
• Update financial model with revised transmission CapEx and OpEx (losses)
• Assess impact of increased transmission losses on project IRR
• Negotiate PPA terms considering lower capacity factor at POI
• Secure financing for transmission infrastructure

DESIGN REFINEMENTS:
• Optimize conductor sizing for 66 kV and 230 kV lines to minimize losses
• Finalize transformer ratings and tap ranges
• Design 66 kV central substation layout at Letpantaw
• Develop single-line diagrams for all substations
• Specify protection relay settings
• Design SCADA/communication architecture